

50Days-coding-challenge

Day-38[Python]

Insurance Premium Calculator

Name:M NITHIN

Problem Statement

Build a Python program that calculates the annual insurance premium for a customer based on their age, policy type, and claim history.

Inputs

Ask the user for:

1. Customer name
2. Age
3. Policy type: health, vehicle, or life
4. Number of claims made in the last year

Base Premium

Policy Type Base Premium

health ₹8,000 vehicle ₹12,000 life ₹10,000

Rules

1. If age is below 25, add **20%** to the premium.
2. If age is between 25 and 45, no age adjustment.
3. If age is above 45, add **15%** to the premium.
4. If claims are more than 2, add **25%** to the premium.
5. If claims are 0, give a **10% discount**.
6. If an invalid policy type is entered, show an error message. **Expected Output**

Display:

Customer Name: Ravi

Policy Type: health

Base Premium: ₹8000 Final Premium: ₹8640

Skills Tested

- Variables

- Input/output
- Conditional statements
- Arithmetic operations
- Basic validation
- Clean code structure

Bonus Task Create a function:

`calculate_premium(age, policy_type, claims)`

The function should return the final premium.

Sample Test Case

Input

Name: Ravi

Age: 23

Policy Type: health

Claims: 0

Calculation

Base premium = 8000

Age loading = 20%

Subtotal = 9600

No claim discount = 10%

Final premium = 8640

Output

Final Premium: ₹8640

```
1 usage
2
3 def calculate_premium(age, policy_type, claims):
4
5     premiums = {
6         "health": 8000,
7         "vehicle": 12000,
8         "life": 10000
9     }
10
11     if policy_type not in premiums:
12         return None
13
14     premium = premiums[policy_type]
15
16     if age < 25:
17         premium += premium * 0.20
18     elif age > 45:
19         premium += premium * 0.15
20
21     if claims > 2:
22         premium += premium * 0.25
23     elif claims == 0:
24         premium -= premium * 0.10
25
26     return premium
27
28 calculate_premium() -> if policy_type not in premiums
```

Run Day38 x

helloworld > Day38.py 10:20 CRLF UTF-8 4 spaces Python 3.13 (helloworld)

```
24     return premium
25
26
27 name = input("Enter Customer Name: ")
28 age = int(input("Enter Age: "))
29 policy_type = input("Enter Policy Type (health/vehicle/life): ").lower()
30 claims = int(input("Enter Number of Claims: "))
31
32
33 final_premium = calculate_premium(age, policy_type, claims)
34
35
36 if final_premium is None:
37     print("Error: Invalid policy type entered.")
38 else:
39     base_premiums = {
40         "health": 8000,
41         "vehicle": 12000,
42         "life": 10000
43     }
44
45     print("\n----- Insurance Premium Details -----")
46     print("Customer Name:", name)
47
48     calculate_premium() -> if policy_type not in premiums
```

Run Day38 x

helloworld > Day38.py 10:20 CRLF UTF-8 4 spaces Python 3.13 (helloworld)

```
32
33 final_premium = calculate_premium(age, policy_type, claims)
34
35
36 if final_premium is None:
37     print("Error: Invalid policy type entered.")
38 else:
39     base_premiums = {
40         "health": 8000,
41         "vehicle": 12000,
42         "life": 10000
43     }
44
45     print("\n----- Insurance Premium Details -----")
46     print("Customer Name:", name)
47     print("Policy Type:", policy_type)
48     print("Base Premium: ₹", base_premiums[policy_type])
49     print("Final Premium: ₹", round(final_premium))
50
51     calculate_premium() -> if policy_type not in premiums
```

Run Day38 x

helloworld > Day38.py 10:20 CRLF UTF-8 4 spaces Python 3.13 (helloworld)

Result:

The screenshot displays the PyCharm IDE interface. The top toolbar shows various icons for file operations, running, and debugging. The main editor window contains a Python script with the following code:

```
44  
45     print("\n----- Insurance Premium Details -----")  
46     print("Customer Name:", name)  
47     print("Policy Type:", policy_type)  
48     print("Base Premium: ₹", base_premiums[policy_type])  
49     print("Final Premium: ₹", round(final_premium))
```

Below the editor, the 'Run' tab is active, showing the command used to execute the script: `C:\Users\NITHIN\PycharmProjects\helloworld\.venv\Scripts\python.exe C:\Users\NITHIN\PycharmProjects\helloworld\Day38.py`. The output of the script is displayed in the console:

```
Enter Customer Name: Nithin  
Enter Age: 24  
Enter Policy Type (health/vehicle/life): health  
Enter Number of Claims: 0  
  
----- Insurance Premium Details -----  
Customer Name: Nithin  
Policy Type: health  
Base Premium: ₹ 8000  
Final Premium: ₹ 8640  
  
Process finished with exit code 0
```

The bottom status bar indicates the current file is `helloworld > Day38.py`, the time is 10:20, and the encoding is UTF-8 with 4 spaces and Python 3.13 (helloworld).